

# Ports of Plague

*Historical Research Sheet · the sources and facts behind the game*

## HOW THE HISTORY WAS BUILT INTO THE GAME

Every historical claim in *Ports of Plague* (on cards, in city pop-ups, in the rules and in the end-of-game summary) is stored as a numbered fact in one file, `data/facts.json`. Each fact lists at least one source and a short supporting quotation. An automatic audit (`npm run audit`) checks that every card, city, route and rule points to a real fact, that every fact has a source, and that the rule numbers match the game. The game uses **85 facts** from **37 sources**, covering all seven required topics:

| TRADE ROUTES | TIMELINE | CITIES & REGIONS | SOCIAL RESPONSES | ECONOMY | MEDICINE | CHURCH |
|--------------|----------|------------------|------------------|---------|----------|--------|
| 11           | 12       | 14               | 13               | 11      | 15       | 9      |

Source types: 13 encyclopedia, 1 scholarly magazine, 8 peer-reviewed article, 1 university publication, 2 library / museum, 1 reputable magazine, 1 official institution, 1 encyclopedia (historical), 1 university resource, 7 primary source, 1 scholarly organization. The plague arrival date for each of the 25 cities on the map comes from these sources; where sources disagree, the game uses one date and shows the other in the city's pop-up.

## HISTORICAL DEBATES AND UNCERTAINTIES

**How many died?** Britannica estimates about 25 million deaths in Europe, roughly one-third of the population, close to the chronicler Froissart's figure. Historian Ole Benedictow, after collecting many local studies, argues for about 60 percent (around 50 million). Death rates also varied greatly by region. [DB-01](#) [CI-06](#)

**Rats or people?** The traditional explanation is fleas carried by black rats. A 2018 study in *PNAS* found that human fleas and body lice fit death records from nine outbreaks better. Ancient DNA has confirmed that the bacterium *Yersinia pestis* caused the Black Death, but scientists still disagree about how it passed from person to person. [ME-10](#) [ME-01](#)

**The siege of Caffa.** The story that attackers threw plague corpses over the walls comes from one writer, Gabriele de' Mussi. A modern analysis finds it plausible, but concludes the siege was only one of several routes out of the Black Sea. [TR-03](#)

**Dates and places.** Sources disagree by weeks or months on when the plague reached some cities (Genoa: November 1347 or January 1348; Melcombe Regis: June or August 1348). Old maps also wrongly showed regions like the Low Countries as "spared" because of gaps in the evidence. [TL-04](#) [TR-08](#) [CI-12](#)

**Chroniclers' numbers.** Medieval writers often exaggerated. For Florence alone they give 60,000 or 100,000 deaths, or three in five people. Numbers like these are treated as rough impressions, not counts. [CI-02](#)

**Anachronisms the game avoids.** Formal quarantine (Ragusa, 1377, 30 days; later 40 days in Venice) came after 1353, so it appears only as "what came next". Earlier, haphazard measures (Venice's ship checks, Pistoia's travel ban, Milan's boarded-up houses) are used instead. The beaked "plague doctor" mask dates from 1619 and is left out. [ME-12](#) [ME-14](#) [ME-15](#) [ME-11](#)

**Persecution.** The game shows the persecution of Jewish communities because it happened, and it states plainly that the well-poisoning accusations were false and the violence unjust. Pope Clement VI condemned the accusations, but his words had little effect far from Avignon. Players can never gain anything from persecution. [SO-07](#) [SO-08](#) [SO-09](#)

Facts marked as debated in the table below: [TR-03](#), [TR-08](#), [TL-04](#), [CI-02](#), [CI-12](#), [ME-10](#), [CH-08](#), [DB-01](#).

## KEY FACTS, THEIR SOURCES, AND WHERE THEY APPEAR IN THE GAME

| ID                                                     | FACT                                                                                                                                 | SOURCE(S) | WHERE IN THE GAME                                            |
|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------|
| Trade routes and how they spread the plague (11 facts) |                                                                                                                                      |           |                                                              |
| <b>TR-01</b>                                           | Caffa (today Feodosiya, in Crimea) was a trading port on the Black Sea founded by the Italian city of Genoa in 1266.<br>1266 · Caffa | [35] [21] | Chronicle card: The Siege of Caffa; City: Caffa; City: Genoa |

| ID                                                  | FACT                                                                                                                                                                                                                                                                                                                                                                                                                             | SOURCE(S) | WHERE IN THE GAME                                                                                                                 |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------|
| TR-02                                               | A Mongol army of the Golden Horde under Khan Janibeg besieged Caffa. In autumn 1346 plague broke out among the besiegers and spread into the town. In spring the Italian merchants fled on their ships, and the disease sailed with them.<br>1346–1347 · Caffa                                                                                                                                                                   | [3] [21]  | Chronicle card: The Siege of Caffa; City: Caffa; Plague arrival: Caffa; Prologue; +2 more                                         |
| TR-03                                               | The notary Gabriele de' Mussi wrote that the attackers catapulted plague corpses into Caffa. A modern scientist finds this plausible, but concludes the siege was probably only one of several ways the plague left the Black Sea region, since ships and caravans from other ports carried it too. <i>Debated: Historians disagree about how reliable de' Mussi's story is and how much the siege mattered.</i><br>1346 · Caffa | [35] [21] | Chronicle card: The Siege of Caffa; City: Caffa                                                                                   |
| TR-04                                               | Large merchant ships linked Venice and Genoa with Constantinople, the Crimea, Alexandria, Tunis, London, and Bruges. At London and Bruges the Italian trade system connected to the ships (cogs) of the German Hanseatic League, which sailed the North Sea and the Baltic.<br>14th century · Mediterranean, North Sea, Baltic                                                                                                   | [3]       | Event card: The Great Network; Action: Ship Goods; Action: Open Trading Post; Action: Propose a Partnership; +22 more             |
| TR-05                                               | Ships averaged about 40 km a day, so plague could jump 600 km by sea in two weeks. Over land it spread far more slowly: up to about 2 km a day on the busiest roads.<br>1347–1353 · Europe                                                                                                                                                                                                                                       | [3]       | Event card: Crowded Ships; Event card: Faster by Sea; Action: Ship Goods; Route: Venice–Florence; +8 more                         |
| TR-06                                               | In early October 1347, twelve Genoese galleys arriving from the east brought the plague into the harbor of Messina in Sicily.<br>October 1347 · Messina                                                                                                                                                                                                                                                                          | [17] [36] | Chronicle card: Twelve Galleys at Messina; City: Messina; Plague arrival: Messina; Round banner: Late 1347; +4 more               |
| TR-07                                               | Italian ships from Caffa reached Constantinople in May 1347 with plague on board; the epidemic broke out there in early July. Ships from Constantinople then carried it to Alexandria and to Mediterranean ports.<br>May–July 1347 · Constantinople                                                                                                                                                                              | [3]       | Chronicle card: Twelve Galleys at Messina; City: Constantinople; Plague arrival: Constantinople; Round banner: Late 1347; +2 more |
| TR-08                                               | A ship brought the plague to Melcombe Regis (part of Weymouth) in Dorset in 1348, and from there it spread across England. Sources disagree on the details: one historian says the ship came from Bordeaux and the outbreak began in June; Britannica says a ship from Calais arrived in August. <i>Debated: Sources give different months and different home ports for the ship.</i><br>1348 · Melcombe Regis                   | [3] [21]  | Chronicle card: A Ship to England; City: Bordeaux; City: Melcombe Regis; Plague arrival: Melcombe Regis; +4 more                  |
| TR-09                                               | Plague reached Bergen in Norway in early July 1349 on a ship from England. Hanseatic ships heading home from Norway then carried it to Baltic and North Sea trading cities in 1349–1350.<br>1349–1350 · Bergen, Baltic                                                                                                                                                                                                           | [3]       | Chronicle card: Hanseatic Ships Sail North; City: Bergen; City: Lübeck; Plague arrival: Bergen; +7 more                           |
| TR-10                                               | The plague did not reach Russia directly from the east. It arrived from the west: it entered the lands of Novgorod in late 1351, broke out in Novgorod itself in August 1352, and ravaged Moscow in 1353.<br>1351–1353 · Novgorod, Moscow                                                                                                                                                                                        | [3]       | Chronicle card: Toward Russia; Chronicle card: Novgorod; Chronicle card: Moscow, 1353; City: Novgorod; +9 more                    |
| TR-11                                               | Right after the Black Death, trade slumped suddenly, but the slump did not last long.<br>1348–1350 · Europe                                                                                                                                                                                                                                                                                                                      | [21]      | Chronicle card: Trade Revives; Event card: Trade Slump; Round banner: Early 1352                                                  |
| <b>Timeline of the plague, 1347–1353 (12 facts)</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |                                                                                                                                   |
| TL-01                                               | Ancient DNA from graves near Lake Issyk-Kul in Kyrgyzstan, whose tombstones mention a "pestilence" in 1338–1339, contains the plague bacterium. Scientists identified it as the ancestor of the strains behind the Black Death, pointing to a source in central Asia.<br>1338–1339 · Central Asia                                                                                                                                | [20]      | Prologue                                                                                                                          |

| ID                                                  | FACT                                                                                                                                                                                                                                                                                                                            | SOURCE(S) | WHERE IN THE GAME                                                                                                |
|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------|
| TL-02                                               | From the Black Sea, the plague reached Sicily in 1347; North Africa, mainland Italy, Spain, and France in 1348; Austria, Hungary, Switzerland, Germany, and the Low Countries in 1349; and Scandinavia and the Baltic in 1350.<br>1347–1350 · Europe                                                                            | [21]      | Chronicle card: The Plague Reaches the Baltic; City: Cologne; City: Vienna; Plague arrival: Tunis; +9 more       |
| TL-03                                               | Plague reached Alexandria around 1 September 1347 and Marseille around the second week of September 1347, both on ships from Constantinople.<br>September 1347 · Alexandria, Marseille                                                                                                                                          | [3]       | City: Alexandria; City: Marseille; Plague arrival: Alexandria; Plague arrival: Marseille; +3 more                |
| TL-04                                               | Britannica dates the plague's arrival to January 1348 for Genoa and Pisa and February 1348 for Venice. Historian Ole Benedictow places Genoa and Venice slightly earlier, in November 1347. <i>Debated: Exact arrival months differ by one to three months between sources.</i><br>Nov. 1347 – Feb. 1348 · Genoa, Pisa, Venice  | [36] [3]  | City: Genoa; City: Pisa; City: Venice; Plague arrival: Genoa; +7 more                                            |
| TL-05                                               | The papal physician Guy de Chauliac wrote that the great mortality began in Avignon in January 1348 and lasted seven months.<br>January–July 1348 · Avignon                                                                                                                                                                     | [13]      | Chronicle card: The Great Mortality at Avignon; City: Avignon; Plague arrival: Avignon; Round banner: Early 1348 |
| TL-06                                               | From Marseille the plague moved up the Rhône valley and along the coast toward Spain. By March 1348 Spain's Mediterranean coast was under attack, and by the end of March Bordeaux had become a new center of spread.<br>March 1348 · Barcelona, Bordeaux                                                                       | [3]       | City: Marseille; City: Barcelona; City: Bordeaux; Plague arrival: Barcelona; +7 more                             |
| TL-07                                               | The plague reached Paris in the summer of 1348. The chronicler Jean de Venette dated its arrival to around August 1348, and it lasted through that year and the next.<br>Summer 1348 · Paris                                                                                                                                    | [15] [3]  | City: Paris; Plague arrival: Paris; Round banner: Late 1348; Route: Avignon–Paris; +2 more                       |
| TL-08                                               | London was infected by about August 1348, and suffered worst between February and May 1349.<br>1348–1349 · London                                                                                                                                                                                                               | [3] [21]  | City: London; Plague arrival: London; Round banner: Late 1348; Route: Melcombe Regis–London                      |
| TL-09                                               | The plague spread mostly in warm seasons. Winter slowed it or stopped it for a while, especially in the Alps and northern Europe.<br>1347–1353 · Europe                                                                                                                                                                         | [3]       | Event card: A Hot Season; Event card: Winter Cold; City: Vienna; City: Bergen                                    |
| TL-10                                               | The Black Death was not the end. Plague returned in 1361–63, 1369–71, 1374–75, 1390, and 1400.<br>1361–1400 · Europe                                                                                                                                                                                                            | [21]      | Chronicle card: Moscow, 1353; Round banner: Early 1353; End-of-game summary                                      |
| TL-11                                               | Iceland and Finland are the only regions known for certain to have escaped the Black Death, because of their small populations and limited contact with other places.<br>1346–1353 · Iceland, Finland                                                                                                                           | [3]       | End-of-game summary                                                                                              |
| TL-12                                               | People at the time called it "the pestilence" or "the great mortality." The name "Black Death" came centuries later, probably from a mistranslation of the Latin word <i>atra</i> , which can mean both "terrible" and "black."<br>later centuries · Europe                                                                     | [3] [13]  | End-of-game summary                                                                                              |
| <b>Major affected cities and regions (14 facts)</b> |                                                                                                                                                                                                                                                                                                                                 |           |                                                                                                                  |
| CI-01                                               | Florence was one of the hardest-hit cities. Historian Ole Benedictow estimates that 60 percent of its people died within a few months of 1348. Boccaccio set his famous book the Decameron in plague-stricken Florence.<br>1348 · Florence                                                                                      | [3] [6]   | Chronicle card: Florence, 1348; City: Florence; Plague arrival: Florence                                         |
| CI-02                                               | Writers of the time gave very different death tolls for Florence: the chronicler Villani said three in every five died, Archbishop Antoninus estimated 60,000, and Boccaccio said 100,000. Numbers like these are rough guesses. <i>Debated: Chroniclers' figures are often exaggerated or inconsistent.</i><br>1348 · Florence | [29]      | Event card: The Chronicler's Tally; City: Florence; Fortune card: The Chronicler's Praise                        |

| ID                          | FACT                                                                                                                                                                                                                                                                                                                                                                                                                     | SOURCE(S) | WHERE IN THE GAME                                                                                                  |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------|
| CI-03                       | Giovanni Villani, a Florentine chronicler who had been a partner in the Peruzzi banking company, died in the plague of 1348.<br>1348 · Florence                                                                                                                                                                                                                                                                          | [27]      | City: Florence                                                                                                     |
| CI-04                       | Siena lost so many people that its plan to enlarge the cathedral was abandoned, and the deaths of painters such as Ambrogio and Pietro Lorenzetti ended the first Sienese school of painting.<br>1348 · Siena (Tuscany)                                                                                                                                                                                                  | [21]      | Event card: Siena's Unfinished Cathedral                                                                           |
| CI-05                       | In Paris, Jean de Venette wrote that more than 500 dead a day were carried from the Hôtel-Dieu hospital to the cemetery, and that many of the nuns who nursed the sick died too.<br>1348–1349 · Paris                                                                                                                                                                                                                    | [15]      | City: Paris                                                                                                        |
| CI-06                       | Death rates varied from place to place. The duchy of Milan, Flanders, and Béarn seem to have escaped comparatively lightly, while Tuscany, Aragon, Catalonia, and Languedoc were very hard-hit.<br>1348–1350 · Europe                                                                                                                                                                                                    | [21]      | Event card: Uneven Losses; City: Pisa; City: Barcelona; City: Bruges; +1 more                                      |
| CI-07                       | Even royalty died. Joan, daughter of King Edward III of England, died at Bordeaux on her way to marry the son of the king of Castile.<br>1348 · Bordeaux                                                                                                                                                                                                                                                                 | [21]      | City: Bordeaux                                                                                                     |
| CI-08                       | The plague struck Tunis in North Africa and killed both parents of the young scholar Ibn Khaldun, who later became a famous historian.<br>1349 · Tunis                                                                                                                                                                                                                                                                   | [28] [21] | Event card: A Scholar's Loss; City: Tunis; Plague arrival: Tunis                                                   |
| CI-09                       | In 1348 the Black Death nearly emptied Ragusa (today Dubrovnik, Croatia), a merchant republic on the Adriatic that depended on grain brought in by ship.<br>1348 · Ragusa                                                                                                                                                                                                                                                | [4]       | City: Ragusa; Plague arrival: Ragusa; Route: Messina–Ragusa; Route: Ragusa–Venice                                  |
| CI-10                       | In 1348–49 the Black Death killed around half of England's population. About 1,000 English villages were emptied or disappeared completely.<br>1348–1349 · England                                                                                                                                                                                                                                                       | [1] [21]  | City: Melcombe Regis; City: London                                                                                 |
| CI-11                       | In Messina the Genoese sailors were driven out, but the disease spread anyway. Terrified people fled to nearby Catania, which refused to let them be buried inside the town.<br>October 1347 · Messina                                                                                                                                                                                                                   | [17]      | Chronicle card: Twelve Galleys at Messina; City: Messina; Rule Book: The Plague; Fortune card: Turned Away at Port |
| CI-12                       | Some older maps show the Low Countries (including Flanders and Bruges) as almost untouched by plague. Newer research shows this came from gaps in the old data: many regions of the Low Countries were hit by the Black Death.<br><i>Debated: How badly the Low Countries, Poland and some other regions were hit is still debated.</i><br>1349 · Bruges, Low Countries                                                  | [19] [21] | City: Bruges; Plague arrival: Bruges                                                                               |
| CI-13                       | The chronicler Jacob von Königshofen recorded that the plague reached Strasbourg in the summer of 1349, months after the city's Jewish community had been murdered. He estimated that about 16,000 people died there.<br>Summer 1349 · Strasbourg                                                                                                                                                                        | [14]      | Chronicle card: Hanseatic Ships Sail North; City: Strasbourg; Plague arrival: Strasbourg; Round banner: Late 1349  |
| DB-01                       | Estimates differ. Britannica gives about 25 million deaths in Europe, around one-third of the population, which matches what the chronicler Froissart said at the time. Historian Ole Benedictow argues for about 60 percent, around 50 million. Regional death rates ranged from one-eighth to two-thirds. <i>Debated: Overall mortality is one of the biggest debates about the Black Death.</i><br>1347–1353 · Europe | [21] [3]  | Event card: The Chronicler's Tally; End-of-game summary; Rule Book: Historical Background                          |
| Social responses (13 facts) |                                                                                                                                                                                                                                                                                                                                                                                                                          |           |                                                                                                                    |

| ID    | FACT                                                                                                                                                                                                                                                                                                                             | SOURCE(S)      | WHERE IN THE GAME                                                                                                                                                         |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SO-01 | Most people believed the plague was God's punishment for their sins. They answered with prayer, processions, and acts of penance, or with fatalism.<br>1347–1353 · Europe                                                                                                                                                        | [3] [30]       | Chronicle card: The Flagellants Arrive;<br>Event card: A Punishment from God?                                                                                             |
| SO-02 | Flagellants marched from town to town in processions with crosses and banners, whipping themselves to atone for sins and calling on people to repent. Their movement grew quickly during the Black Death. Members vowed to take part for 33 and a half days.<br>1348–1349 · Germany, Low Countries, France                       | [26] [33] [15] | Chronicle card: The Flagellants Arrive;<br>City: Cologne; Rule Book: Glossary                                                                                             |
| SO-03 | Pope Clement VI condemned the flagellant movement and banned its public processions in letters dated 20 October 1349. After that the movement shrank quickly.<br>20 October 1349 · Avignon                                                                                                                                       | [33] [15]      | Chronicle card: The Pope Bans the Flagellants                                                                                                                             |
| SO-04 | Many people who could afford it fled the cities for the countryside. Boccaccio's Decameron begins with a group of young Florentines leaving the city, and the physician Guy de Chauliac wrote that the best protection was to flee before the plague arrived.<br>1348 · Florence, Avignon                                        | [6] [13]       | Chronicle card: Florence, 1348; Event card: Flight to the Countryside; Event card: A Doctor Who Stayed; Action: Move Family; +2 more                                      |
| SO-05 | Some families shut themselves inside houses where no one was sick and lived apart from everyone else.<br>1348 · Florence                                                                                                                                                                                                         | [6]            | Event card: Behind Closed Doors; Action: Prepare Household; Rule Book: Actions                                                                                            |
| SO-06 | Fear broke normal bonds. Writers described brothers abandoning brothers and even parents abandoning their children. Funerals became rushed or stopped altogether.<br>1348 · Florence, Avignon                                                                                                                                    | [6] [13] [29]  | Event card: Families Torn Apart; Fortune card: Servants Flee                                                                                                              |
| SO-07 | Across Europe, Jews were falsely accused of causing the plague by poisoning wells. The accusation was untrue: Jews were dying of the plague too. Beginning in September 1348, Jews were tortured into false "confessions."<br>1348–1349 · Europe                                                                                 | [2] [15] [13]  | Chronicle card: A False Accusation, and a Pope's Reply; Chronicle card: Strasbourg, February 1349; Rule Book: Historical Background; Rule Book: Chronicle and Event Cards |
| SO-08 | In 1348 Pope Clement VI issued papal bulls (official letters) that rejected the poisoning accusation and called for Jews to be protected. He also welcomed Jews in Avignon. Elsewhere his appeals had little effect, and the massacres continued.<br>1348 · Avignon                                                              | [12] [2] [23]  | Chronicle card: A False Accusation, and a Pope's Reply; Rule Book: Chronicle and Event Cards                                                                              |
| SO-09 | On 14 February 1349, before the plague had even reached Strasbourg, the city's Jewish community was murdered. The city council, which had refused to act against the Jews, was overthrown first. The chronicler Königshofen admitted that debts owed to Jewish moneylenders were a real motive.<br>14 February 1349 · Strasbourg | [14] [2]       | Chronicle card: Strasbourg, February 1349; City: Strasbourg; Rule Book: Chronicle and Event Cards                                                                         |
| SO-10 | Some leaders tried to protect Jewish communities, such as the old city council of Strasbourg and the leading citizens of Basel, but mobs overruled them.<br>1349 · Strasbourg, Basel                                                                                                                                             | [2] [14]       | Chronicle card: Strasbourg, February 1349; City: Strasbourg                                                                                                               |
| SO-11 | Towns posted guards at their gates and turned away strangers. Pistoia in Tuscany banned its people from traveling to or from nearby Pisa and Lucca.<br>1348 · Pistoia, Avignon                                                                                                                                                   | [13] [9]       | Chronicle card: Cities Guard Their Gates; Event card: Guards at the Gates; Event card: Cargo Seized at the Gate; Action: Close Your Gates; +3 more                        |
| SO-12 | After the epidemic ended, the survivors married, and Jean de Venette noticed an unusually large number of births.<br>1349–1350 · France                                                                                                                                                                                          | [15]           | Chronicle card: Survivors Marry; Action: Arrange a Marriage; Round banner: Late 1350; Rule Book: Actions; +1 more                                                         |

| ID                                                     | FACT                                                                                                                                                                                                                                                | SOURCE(S)     | WHERE IN THE GAME                                                                                                                |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>SO-13</b>                                           | With so many workers dead, the survivors gained bargaining power. This brought a new flexibility to what had been a very rigid social order.<br>after 1350 · Europe                                                                                 | [21]          | Chronicle card: Lords Must Pay; Event card: New Opportunities                                                                    |
| <b>Economic consequences (11 facts)</b>                |                                                                                                                                                                                                                                                     |               |                                                                                                                                  |
| <b>EC-01</b>                                           | Italy's economy was already struggling before the plague arrived. The great Florentine banks failed: the Peruzzi in 1343, the Acciaiuoli in 1345, and the Bardi in 1346. There were also famines in 1346–47.<br>1343–1347 · Florence, Italy         | [36]          | Event card: Memories of the Bank Crash; Event card: An Old Loan; Action: Take a Loan; City: Florence; +2 more                    |
| <b>EC-02</b>                                           | So many laborers died that much less land could be farmed, which ruined many landowners. To keep tenants, lords began paying wages or charging money rents instead of demanding unpaid labor.<br>after 1349 · Europe                                | [21]          | Chronicle card: Lords Must Pay; Event card: Fields Go Unplowed; Action: Buy Abandoned Land; Rule Book: Actions; +1 more          |
| <b>EC-03</b>                                           | Because workers were scarce, wages rose for both craftsmen and peasants.<br>after 1349 · Europe                                                                                                                                                     | [21] [29]     | Chronicle card: The Ordinance of Labourers; Event card: Wages Rise; Action: Buy Abandoned Land; Rule Book: Actions; +1 more      |
| <b>EC-04</b>                                           | Prices rose too. Jean de Venette complained that after the plague everything cost twice as much: tools, food, goods, and hired workers.<br>after 1349 · France                                                                                      | [15] [29]     | Chronicle card: Everything Twice as Dear; Event card: Prices Soar; Round banner: Late 1350; Rule Book: The Plague; +1 more       |
| <b>EC-05</b>                                           | On 18 June 1349, while the plague was still raging, the English Crown issued the Ordinance of Labourers. It capped wages at pre-plague levels and required able-bodied people under 60 to work.<br>18 June 1349 · England                           | [1] [31]      | Chronicle card: The Ordinance of Labourers; City: London; Rule Book: Historical Background; Rule Book: Chronicle and Event Cards |
| <b>EC-06</b>                                           | In February 1351, at the first Parliament after the plague, the Statute of Labourers strengthened these rules. It complained that workers refused to work unless paid two or three times the old wages.<br>February 1351 · England                  | [1] [31]      | Chronicle card: The Statute of Labourers; City: London; Round banner: Early 1351; Fortune card: Workers Demand Double            |
| <b>EC-07</b>                                           | The labor laws soon failed to hold wages down. They caused lasting resentment, which fed into the grievances behind the Peasants' Revolt of 1381.<br>1351–1381 · England                                                                            | [1] [7]       | Chronicle card: The Statute of Labourers; End-of-game summary; Rule Book: Chronicle and Event Cards                              |
| <b>EC-08</b>                                           | Western Europe's population did not return to its pre-1348 level until the early 1500s.<br>1348–1500s · Western Europe                                                                                                                              | [21]          | Chronicle card: Moscow, 1353; End-of-game summary                                                                                |
| <b>EC-09</b>                                           | For a short time the plague brought a pause in wars.<br>1348–1350 · Europe                                                                                                                                                                          | [21]          | Event card: Wars Pause                                                                                                           |
| <b>EC-10</b>                                           | So many people died that survivors inherited their property, sometimes more land than they wanted. For roughly a century afterward, wealth was spread more evenly, as the richest lost part of their share to everyone else.<br>after 1348 · Europe | [10]          | Event card: An Unexpected Will; Action: Buy Abandoned Land; Rule Book: The Plague; Fortune card: A Cousin's Will                 |
| <b>EC-11</b>                                           | The gold florin, first minted in Florence in 1252, won fame across Europe for its reliable weight and purity and was widely imitated.<br>1252 · Florence                                                                                            | [25]          | Action: Take a Loan; Rule Book: Glossary; Fortune card: A Letter of Credit                                                       |
| <b>Medical understanding and treatments (15 facts)</b> |                                                                                                                                                                                                                                                     |               |                                                                                                                                  |
| <b>ME-01</b>                                           | Today we know the Black Death was plague, caused by the bacterium <i>Yersinia pestis</i> . DNA taken from medieval skeletons has confirmed this. No one at the time knew bacteria existed.<br>modern research                                       | [21] [20] [3] | Action: Consult Physician; End-of-game summary                                                                                   |



| ID    | FACT                                                                                                                                                                                                                                                                                                                    | SOURCE(S)     | WHERE IN THE GAME                                                                                                                                  |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| ME-02 | Bubonic plague causes painful swellings called buboes in the groin, armpits, or neck. If the infection reaches the lungs it becomes pneumonic plague, a more severe form that can spread through coughing.                                                                                                              | [22] [3]      | Chronicle card: The Great Mortality at Avignon; Physician remedy: Figs and onions; Fortune card: Illness in the House                              |
| ME-03 | In Avignon, Guy de Chauillac described two forms. For the first two months, victims had fever and coughed up blood, and died within three days. After that, victims had swellings in the armpits and groin and died within five days.<br>1348 · Avignon                                                                 | [13]          | Chronicle card: The Great Mortality at Avignon; City: Avignon; Fortune card: Illness in the House                                                  |
| ME-04 | In October 1348 the medical faculty of the University of Paris reported that the plague was caused by a conjunction of three planets on 20 March 1345, which had supposedly corrupted the air.<br>October 1348 · Paris                                                                                                  | [30]          | Chronicle card: The Paris Report; Event card: Astrologers' Advice; Physician remedy: Fires to purify the air; Physician remedy: The stars; +2 more |
| ME-05 | Medieval doctors believed illness came from "corrupt air" (the miasma theory) and from an imbalance of the body's humors (fluids). Guy de Chauillac wrote that the planets had turned the body's humors "venomous."<br>1348 · Paris, Avignon                                                                            | [30] [13]     | Chronicle card: The Paris Report; Event card: A Hot Season; Physician remedy: Bloodletting; Rule Book: Glossary; +1 more                           |
| ME-06 | Doctors tried bloodletting, purging pills of aloe, fires to "purify" the air, a medicine called theriac, sweet-smelling herbs, and poultices of cooked figs and onions on the swellings. None of these could cure plague.<br>1348 · Avignon                                                                             | [13] [22]     | Event card: A Remedy Seller; Physician remedy: Bloodletting; Physician remedy: Pills of aloe; Physician remedy: Fires to purify the air; +4 more   |
| ME-07 | People in Florence walked around holding flowers, fragrant herbs, or spices to their noses, hoping good smells would protect them from bad air.<br>1348 · Florence                                                                                                                                                      | [6]           | Chronicle card: Florence, 1348; Physician remedy: Sweet-smelling herbs; City: Florence                                                             |
| ME-08 | Both Boccaccio and the physician Guy de Chauillac admitted that medicine could do almost nothing against the plague. Today plague is treated with antibiotics.<br>1348 · Florence, Avignon                                                                                                                              | [6] [13] [22] | Event card: A Remedy Seller; Action: Consult Physician; Rule Book: Historical Background                                                           |
| ME-09 | Guy de Chauillac stayed in Avignon to avoid a bad reputation. Near the end of the outbreak he caught the plague himself, was ill for about six weeks, and survived.<br>1348 · Avignon                                                                                                                                   | [13]          | Event card: A Doctor Who Stayed; City: Avignon; Fortune card: A Doctor Who Stays                                                                   |
| ME-10 | Historians still debate how plague spread from person to person. The traditional view blames fleas on black rats. A 2018 study found that human fleas and body lice fit the death records of nine European outbreaks better. <i>Debated: Rat fleas versus human fleas and lice is an open scientific debate.</i>        | [3] [8]       | Event card: Crowded Ships; End-of-game summary; Fortune card: A Clean Hold                                                                         |
| ME-11 | The famous beaked "plague doctor" costume did not exist during the Black Death. It was invented by the French physician Charles de Lorme in the 1600s, almost 300 years later.<br>1619 · Paris                                                                                                                          | [18] [5]      | Event card: Myth Buster: The Beaked Mask                                                                                                           |
| ME-12 | During the Black Death some Italian cities tried early health measures. By 1348 Venice required incoming ships to wait before sailors could enter the city, and in Milan the ruler Luchino Visconti ordered houses with sick people boarded up. These measures were haphazard and limited.<br>1347–1353 · Venice, Milan | [9]           | Chronicle card: Cities Guard Their Gates; Action: Prepare Household; City: Venice; Rule Book: The Plague; +1 more                                  |
| ME-13 | England had no coordinated public-health response to the Black Death, not even restrictions on travel or isolation of the sick. The government's main concern was keeping order.<br>1348–1349 · England                                                                                                                 | [1]           | Chronicle card: A Ship to England                                                                                                                  |
| ME-14 | Formal quarantine came later. On 27 July 1377, after the plague had kept returning, Ragusa ordered travelers from plague areas to spend 30 days in isolation on the island of Mrkan or in the town of Cavtat before entering.<br>27 July 1377 · Ragusa                                                                  | [4] [11]      | City: Ragusa; End-of-game summary; Rule Book: The Plague; Rule Book: Glossary                                                                      |

| ID                                                          | FACT                                                                                                                                                                                                                                                                                 | SOURCE(S)             | WHERE IN THE GAME                                                                                                                              |
|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| ME-15                                                       | The 30-day wait was later extended to 40 days, which Venice also adopted. Our word "quarantine" comes from quaranta, Italian for forty. Venice opened the first permanent plague hospital (lazaretto) in 1423.<br>after 1377; 1423 · Venice                                          | [4] [34] [32]<br>[11] | City: Venice; End-of-game summary                                                                                                              |
| <b>The Catholic Church and religious practice (9 facts)</b> |                                                                                                                                                                                                                                                                                      |                       |                                                                                                                                                |
| CH-01                                                       | During the Black Death the popes lived in Avignon, not Rome (1309–1377). Clement VI was pope from 1342 until his death in Avignon in 1352.<br>1309–1377 · Avignon                                                                                                                    | [23]                  | Chronicle card: The Great Mortality at Avignon; Chronicle card: Death of Clement VI; Event card: A Letter from Avignon; City: Avignon; +1 more |
| CH-02                                                       | About one-fourth of Pope Clement VI's staff at Avignon died in the Black Death.<br>1348 · Avignon                                                                                                                                                                                    | [23] [21]             | Chronicle card: The Great Mortality at Avignon; Event card: Petitions to Avignon; City: Avignon                                                |
| CH-03                                                       | Priests caught the disease while caring for the sick and dying. In English dioceses about 45–50 percent of the clergy died, and monasteries suffered some of the highest death rates of all.<br>1348–1349 · England, Europe                                                          | [37] [21]             | Chronicle card: Confess to Anyone; Event card: The Clergy Hit Hard; Action: Charity & Piety                                                    |
| CH-04                                                       | Canterbury, England's most important church post, lost two archbishops one after the other: John de Stratford and Thomas Bradwardine.<br>1348–1349 · Canterbury (England)                                                                                                            | [21]                  | Event card: Two Archbishops in a Row; City: London                                                                                             |
| CH-05                                                       | So many priests died that in 1349 Ralph of Shrewsbury, Bishop of Bath and Wells, told his people that if no priest could be found, the dying could confess their sins to any layperson, "even to a woman."<br>1349 · England                                                         | [37]                  | Chronicle card: Confess to Anyone                                                                                                              |
| CH-06                                                       | Faith in the Church weakened, partly because so many clergy had died. The Church lost some of its monopoly over saving souls as people turned to mysticism. North of the Alps, art and poetry became preoccupied with death.<br>after 1348 · Europe                                  | [21] [29]             | Event card: Faith Shaken                                                                                                                       |
| CH-07                                                       | After the Black Death, more ordinary people practiced religion on their own, using prayer books such as Books of Hours and asking for religious writings in their own language instead of Latin. Criticism of clergy also grew.<br>after 1350 · England                              | [37]                  | Event card: Religion at Home; Action: Charity & Piety; Fortune card: A Confraternity Invites You                                               |
| CH-08                                                       | In 1343 Clement VI had declared a Holy Year (Jubilee) for 1350. Despite the plague and an earthquake in Rome in 1349, huge crowds of pilgrims came to Rome. The Vatican reports more than one and a half million. <i>Debated: Medieval crowd sizes are estimates.</i><br>1350 · Rome | [16]                  | Chronicle card: The Jubilee of 1350; Round banner: Early 1350; Rule Book: Chronicle and Event Cards; Fortune card: A Pilgrim's Blessing        |
| CH-09                                                       | A papal bull is an official document from the pope. Metal seals were called bulls, and the name passed to the papal documents that carried them.                                                                                                                                     | [24]                  | Rule Book: Glossary                                                                                                                            |

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